Ram Certificates for Analytic Caustic Collars: M282

Exact Circle Preconditioning, Rational Resonances,
Analytic-Scale Budgets, and Quantitative Physical Geometry

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Technical Report Assisted by AI Models
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Technical Report TR-Ram-OpenAI-722-M282

October 7, 2026

Abstract

The public manuscript *Continuous Phase Foliations Create Analytic Caustic Collars* claims that a complete continuous invariant foliation at the grazing end of a smooth positively curved billiard produces an analytic boundary and a jointly analytic physical caustic collar; its ellipse conclusion then uses a companion rigidity theorem. We reconstruct that dependency chain, distinguishing its topological, analytic, resonant, and geometric obligations. Actual local Ram sources supply domain-aware localization, physical Gram metrics, complete residuals, and explicit inverse conditions, rather than an independent global rigidity theorem. We prove an exact circle multiplier with optimal weighted-space inverse, a centered rational primitive with sharp supremum norm, an exact reciprocal-orbit-average flux inverse on fixed holomorphic strips, a real coercive rational inverse beyond the near-constant coefficient regime, and explicit double-exponential analytic-scale stopping budgets. We also prove quantitative support-collar and area certificates, including their stability under fully bounded representation errors. Exact finite checks support the algebra; the proofs establish the displayed infinite statements. These are concrete local or conditional refinements, not a certification of the companion's global continuation argument, a new unrestricted Birkhoff theorem, or a historical-priority claim.

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1 Target, scope, and principal conclusions

Let $\Omega \subset \mathbb{R}^2$ be bounded and strictly convex, with a regular C^∞ simple closed boundary of positive curvature. Its billiard map acts on $M = (\mathbb{R}/L\mathbb{Z}) \times (0, \pi)$, with arclength s and outgoing tangent angle δ. The actual hypothesis in M282 is the existence of a T-invariant open annulus U containing a full strip $(\mathbb{R}/L\mathbb{Z}) \times (0, \eta)$ for some $\eta > 0$, and a homeomorphism $H : \mathbb{S}^1 \times (0, 1) \to U$ whose essential simple closed leaves are *individually* invariant. No differentiability across leaves, symmetry, quantitative closeness to an ellipse, or fixed lower bound on η is assumed.

The source's `thm:collar-regularity` concludes real analyticity of the boundary and a jointly real-analytic parametrization $\Gamma : \mathbb{T} \times [0, \lambda_0] \to \mathbb{R}^2$ extending to a neighborhood, with positive leaves smooth strictly convex caustics, and with $\mathbb{T} \times [0, \lambda_0)$ mapped homeomorphically onto a relatively open collar in $\overline{\Omega}$. Its `thm:main` concludes

$$\Omega = \{x : (x - c)^T Q(x - c) < 1\}, \qquad Q = Q^T > 0,$$

by invoking the physical-collar theorem of the companion [2]. The public source and its commit-pinned location are given in Section 11; the private corpus is separately identified.

This is direct core applicability, consistent with the catalogue's L5 assessment: the geometry and operators match TR-item-336's literal target. It is not a claim that the corpus already contains a proof of the source's theorem. Conversely, the private report's own restricted L3 claim does not make this core match peripheral.

Result in this report	Exact meaning and remaining condition
Circle row operator, Theorem 4.1	Exact diagonal derivative at the normalized circle, on the *whole* high-frequency sequence space. Perturbation away from it needs the source's validated row bounds.
Analytic-scale budget, Theorem 5.1	Explicit ball and tail conditions for an already justified shrinking-strip Newton step; not an unrestricted same-space implicit-function theorem.
Rational inverses, Theorems 6.1, 6.3, and 6.5	The whole rational resonant subspace is removed. Strip-supremum and real L^2 results have different hypotheses and are not interchanged.
Physical residual, Proposition 7.1	A finite Gram residual plus its omitted component bounds the error for the original rational operator when the coercivity gate holds.
Quantitative collar, Theorem 8.1	Support positivity, curvature, inward speed, nesting, area and inverse-coordinate bounds. Caustic status additionally requires exact billiard invariance.
Ellipse conclusion	An imported source claim relying on the companion's global rigidity theorem, not independently established here.

2 Source inventory and dependency reconstruction

The pinned build has *five section files total*, including the introduction: `introduction.tex`, `geometry.tex`, `analytic.tex`, `joint.tex`, and `caustics.tex`. All five, the main file, bibliography and figure source were read in full. There is no sixth section in its input list. Reading is distinct from independent certification of every estimate or every cited theorem.

2.1 From the phase foliation to periodic graphs

Put $n_\theta = (\cos\theta, \sin\theta)$ and $e_\theta = (-\sin\theta, \cos\theta)$. After translating the origin into the table, its support function h satisfies

$$h > 0, \quad r = h + h'' > 0, \quad \gamma = hn_\theta + h'e_\theta, \quad \gamma' = re_\theta. \tag{2.1}$$

An outgoing oriented line with velocity e_θ has equation $z \cdot n_\theta = p$. The complete line-state domain is $-h(\theta+\pi) < p < h(\theta)$. The generating action and momenta are

$$S(\theta, \theta') = 2h((\theta+\theta')/2)\sin((\theta'-\theta)/2), \qquad p = -S_1, \quad p' = S_2. \tag{2.2}$$

Writing $\theta' = \theta + 2a$, the source obtains

$$p = h(\theta+a)\cos a - h'(\theta+a)\sin a, \qquad \partial_a p = -r(\theta+a)\sin a < 0.$$

The inverse formula gives the opposite twist sign: $\partial_p\theta' < 0$ and $\partial_p\theta_{-1} > 0$, at fixed current angle. The map preserves $\mathrm{d}p \wedge \mathrm{d}\theta$ and a finite full-support area measure. These are source `geo:twists` and `geo:generating`; the identities can also be checked directly by differentiating (2.2).

The detailed proof of `prop:graph-collar` is not simply an invocation of a graph theorem. For lifted leaf parameters G_b it uses $|G_b^j(\alpha) - \alpha - j\tau(b)| \le 2\pi$. For an ordered pair of states let x_j be their angular separation. The two twists imply $x_j = 0 \Rightarrow x_{j-1}x_{j+1} < 0$. Source `geo:barrier` constructs a real argument A_j of (x_j, x_{j+1}), with one-way barriers $\pi/2 + \pi m$. A sign change in x_j forbids recurrence of this argument.

On an interval of equal translation numbers, the simultaneous cover of the product annulus has bounded relative parameter displacement:

$$|(G_{b_2}^j(\beta) - G_{b_1}^j(\alpha)) - (\beta - \alpha)| \le 4\pi.$$

The saturation of a bounded-relative-displacement open set therefore has finite lifted product area. Its recurrence contradicts the barrier if a vertical same-leaf pair exists. This finite-measure restriction matters: recurrence on the entire infinite cover would not follow. No absolute continuity of the supplied foliation chart is assumed.

If translation numbers are not constant on an interval, continuity and the strict signs of rational iterate displacements yield a leaf with $G_b^q = \mathrm{id} + 2\pi l$ by approaching a rational value from both sides. Such wholly periodic leaves are graphs by the barrier. Graph leaves are dense; limiting nondecrease and the two twists remove vertical segments. Disjointness orders all graph functions $g_b(\theta)$, so the induced angular maps are strictly ordered and their translation numbers nondecreasing. A rational plateau would make T^q the identity almost everywhere there by recurrence of zero-rotation circle maps, hence everywhere by continuity and full support, contradicting strict order. This proves the source's uniqueness and whole-leaf periodicity at every sufficiently small rational rotation. The full neighboring foliation is essential to this chain.

2.2 Stationary closure and high-frequency equations

For a chain with positive sufficiently small angular steps, $\theta_q = \theta_0 + 2\pi l$, interior stationarity means $S_2(\theta_{j-1}, \theta_j) + S_1(\theta_j, \theta_{j+1}) = 0$. It describes successive billiard states. Both endpoints lie on the same invariant graph, so angle return forces $p_q = p_0$. For a C^1 family of these fixed-winding chains,

$$\frac{\mathrm{d}}{\mathrm{d}x}\sum_{j=0}^{q-1} S(\theta_j, \theta_{j+1}) = -p_0\theta_0' + p_q\theta_q' = 0. \tag{2.3}$$

This is `lem:stationary-closure`; it does not assume endpoint stationarity before proving momentum return.

The Lazutkin-type normalization is

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = kr(\theta)^{1/3}, \quad k\int_0^{2\pi} r^{1/3}\,\mathrm{d}\theta = 2\pi, \quad \theta = \phi(x), \quad u = \log\phi', \quad c_* = k^{-3}. \tag{2.4}$$

Thus $r(\phi) = c_* e^{-3u}$. Subtracting the endpoint primitive $\int h$ from S and dividing by c_* gives an action D_u with unchanged stationary chains. The useful arbitrary-input definition is

$$\begin{aligned} D_f(X,Y) = &-\int_X^{\mu} [1-\cos(\phi_f(z)-\phi_f(X))]e^{-2f(z)}\,\mathrm{d}z \\ &-\int_{\mu}^{Y} [1-\cos(\phi_f(Y)-\phi_f(z))]e^{-2f(z)}\,\mathrm{d}z, \end{aligned} \tag{2.5}$$

where $\phi_f' = e^f$ and $\phi_f(\mu) = (\phi_f(X)+\phi_f(Y))/2$. For arbitrary periodic trial f no normalization of $\int e^f$ is imposed. This removes an otherwise artificial constraint on analytic approximants. The universal expansion is $D_f(x,x+b) = -b^3/24 + O(b^5)$.

The companion's stationary-mesh lemma gives $X_j = x + 2\pi j/q + w_j$, $w_0 = w_q = 0$, with scaled bounds $q^2 \max\|w_j\|_{C^p} + q^3 \max\|w_{j+1}-w_j\|_{C^p} \le R$ on a row-dependent smaller strip. Define

$$W_q(x;f) = \sum_j D_f(X_j, X_{j+1}), \qquad C_{\pm q}(f) = -\frac{q^2}{6\pi}[W_q(\cdot;f)]_{\pm q}. \tag{2.6}$$

For the actual u, (2.3) implies $C_n(u) = 0$ at every sufficiently high frequency. The companion's arbitrary-input row estimate is the substantially stronger analytic input

$$\begin{aligned} (\,\mathrm{d}C_n)_f(v) &= [b_n(f)v]_n, \\ \|b_n(f)-1\|_{m,(1-K/n^2)\sigma} &\le K/|n|, \\ \|b_n(f)-b_n(\widetilde{f})\|_{m,(1-K/n^2)\sigma} &\le (K/|n|)\|f-\widetilde{f}\|_{r_0,\sigma}, \quad r_0 = m+15. \end{aligned} \tag{2.7}$$

Its constants require a fixed finite input bound M and sufficiently large $|n|$. The source is not claiming these bounds for arbitrary rough Wiener inputs. Exact companion locators are recorded in Section 11.

2.3 Analytic inverse and real uniqueness

For $a \ge 0$, let

$$|v|_{\sigma,a} = \sum_{n\in\mathbb{Z}} (1+|n|)^a |v_n| e^{\sigma|n|}.$$

Source `ana:wiener` integrates by parts in (2.7) and sums *all* weighted matrix columns. With $\alpha = K/n^2$,

$$|n| - |j| - (1-\alpha)|n-j| \le \alpha|n| = K/|n|. \tag{2.8}$$

Indeed, subtract $(1-\alpha)|n-j|$ from the triangle inequality term, then bound $\alpha(|n|-|j|) \le \alpha|n|$. This absorbs the row-dependent strip loss and yields a bounded $A_H(f) = I + R_H(f)$ on the entire high-mode Wiener space, with $\|R_H\| \le C/N$. For polynomial weights one requires $m > a+1$. A sufficiently large N gives the quoted inverse bound 2.

The nonlinear proof still uses smaller strips to pay for r_0 derivatives. It starts from the actual smooth Fourier truncation f_0 at cutoff P, width $1/P$. The low rows of $C(f_0)$ use $C(u) = 0$ and a real derivative comparison; the infinitely many high rows use a separate higher-order bound along λf_0 and mesh compatibility. This is why the source can obtain $|C(f_0)|_{1/P} = O(P^{-a})$ for each fixed a, not merely small real residuals. Newton correction on $\sigma_j = (1 + 2^{-j})/(2P)$ gives a holomorphic f_*. Finally the real segment from u to f_* lies in the input ball and

$$0 = C(f_*) - C(u) = v + \int_0^1 R(u + \lambda v)v \, \mathrm{d}\lambda, \qquad v = f_* - u = \Pi_H v.$$

The real operator bound $1/2$ forces $v = 0$. This is a route from smooth data to analyticity using exact equations and uniqueness, not the invalid assertion that a smooth Taylor series already converges. Section 5 makes its stopping budget more explicit, conditional on the analytic estimates just stated.

2.4 Joint parameters: two divisions, not one

Put $T_z f(y, t) = f(y + z, t)$ and $\partial_t^{\mathrm{diff}} f = (T_{t/2} f - T_{-t/2} f)/t$; below write this difference operator as ∂, reserving dots for parameter jets. For a candidate V the source defines

$$\begin{aligned} \mathcal{R}(V) &= D_1(V, T_t V) + D_2(T_{-t} V, V), \\ E(V) &= t^{-3} V' \mathcal{R}(V), \\ a(V) &= t^{-1} D_{12}(T_{-t/2} V, T_{t/2} V)(T_{-t/2} V')(T_{t/2} V'). \end{aligned} \tag{2.9}$$

The apparent poles at zero are removable. The exact identities are

$$E(y, 0) = \frac{(V')^2 V''}{4}, \quad a(y, 0) = \frac{(V')^3}{4}, \quad E_V(V'w) = \partial(a\partial w) + (Ew)', \quad \langle E \rangle = 0. \tag{2.10}$$

The mean-zero identity follows by differentiating the mean action under translation of y. At an exact solution and a nonzero real t_0, an *assigned* jet $\dot{V} = V'z$ gives

$$\dot{E} = \partial(a\partial z + a/t_0). \tag{2.11}$$

The explicit shifts must be differentiated as well as V.

Source `joint:compatibility` first corrects a rational fixed-endpoint chain, using a Dirichlet flux inverse, then uses graph closure and a finite orbit average to construct an exact rational parametrization. A separate cyclic inverse adjusts its assigned first jet. No derivative of the original continuous transverse foliation is used. The corrections w and $\dot{w}$ give both the value and first jet of $E + \partial(a\partial w)$ at order $O_{\mathrm{p}}(\varepsilon^2)$.

In Fourier mode $n \neq 0$, ∂ has divisor $d_n(t) = 2\mathrm{i} \sin(nt/2)/t$, with removable $d_n(0) = \mathrm{i}n$ and simple nonzero zeros $t_j = 2\pi j/n$. The first buffered sinc division approximately solves $\partial H = -E$. One then sets $b = a^{-1}(H + \lambda)$ with $\lambda = -\langle a^{-1} H \rangle / \langle a^{-1} \rangle$, and approximately solves $\partial v = b$. At a resonant center the value equation $d_n(H_n - g_n) =$ quadratic error says nothing about $H_n - g_n$. Its first jet does, because $|d_n'(t_j)| = |n|/|t_j|$. The required second compatibility is therefore a genuine additional condition. The interpolation buffers cover *every* possible pole in the smaller disk; omitted modes are charged by $e^{-N\delta}$.

The slanted domains $\mathcal{D}^\nu(s, d) = \{|t| < d, \ |\Im y| < s - \nu|\Im t|\}$ track shifts: $1 \to 1/2 \to 0$ across the two divisions. A fixed loss exponent B_0 is chosen before the finite formal initialization order. Only after that does the source shrink d_0. This ordering, summable domain losses, and candidate-ball closure are necessary for the actual analytic conclusion. A finite formal recursion alone proves none of them.

2.5 The last geometric and external gates

The exact jointly analytic even solution gives $\Theta = \phi(V)$ and $p = -S_1(\Theta(y,t), \Theta(y+t,t))$. Their line states obey the *actual* billiard map and are even in t. Passing to $\lambda = t^2$ and inverting the angular coordinate gives

$$g(\theta, \lambda) = h(\theta) - c(\theta)\lambda + O(\lambda^2), \qquad c(\theta) = \frac{r(\theta)^{1/3}}{8k^2} > 0. \tag{2.12}$$

The explicit form of c follows from (2.4) and the source's $c = r\Theta_y^2/8$ at $t = 0$. Curvature and inward support motion produce the physical collar; Section 8 proves quantitative versions. The companion's Definition 1.1 and Theorem 1.2 then match the constructed collar. Its later global continuation, shape and sign-test arguments remain an imported obligation here.

3 Actual Ram mechanisms and their domains

The private sources below were read directly, not inferred from a map. Their classical ingredients are acknowledged. Their value here is an exact representation and error accounting tied to the operator at hand.

3.1 Localized action derivatives

Revised TR117 [12], `eq:RLT/eq:IRLT` and "Equivalence of global and Ram forms," gives the change of kernel coordinates $h(s,\alpha) = K(s+\alpha, s)$ and $Tf(x) = \int h(x-\alpha,\alpha) f(x-\alpha)\,\mathrm{d}\alpha$ with the transformed limits. This is kernel localization, not inversion of T. For one positive cell of length b, the circle variation derived below is precisely

$$L_b v(x) = \int_0^b K_b(s) v(x+s)\,\mathrm{d}s = \int_{-b}^0 K_b(-\alpha) v(x-\alpha)\,\mathrm{d}\alpha, \tag{3.1}$$

where $K_b(s) = 2 + \cos(b/2) - 3\cos(\min(s, b-s))$. The kernel is signed; no probability normalization is assumed.

Volume 1 [13], `thm:ram-local`, applies with its actual segment hypotheses. If $v \in C^{J+1}$ on every contributing segment, finite Taylor expansion gives

$$L_b v(x) = \sum_{j=0}^{J} \frac{\mu_j(b)}{j!} v^{(j)}(x) + r_J(x), \qquad \mu_j(b) = \int_0^b K_b(s) s^j\,\mathrm{d}s, \tag{3.2}$$

with

$$|r_J(x)| \le \frac{\|v^{(J+1)}\|_\infty}{(J+1)!} \int_0^b |K_b(s)| s^{J+1}\,\mathrm{d}s.$$

This follows by integrating the ordinary Taylor remainder and is exact on the stated polynomial class. It is not exact for a general smooth periodic input. For $v = e^{\mathrm{i}qx}$ and $b = 2\pi/q$, the oscillation over a cell is not small: $qb = 2\pi$. The exact Fourier integral in Theorem 4.1 is consequently more useful than a low-order uncontrolled jet approximation. This is a concrete choice between two representations of the same Ram-localized operator.

3.2 Resonance, norms and residuals

TR-item-336's [11] "Rotation cohomology: exact rational range and sharp norm" proves the full resonant projection and inverse for $I - T_{2\pi p/q}$. Its `eq:fulltail` and `eq:actiontail` explicitly charge omitted Fourier coefficients. Its `thm:exclusion` proves a continuous-graph exclusion radius from the true nonlinear billiard residual. These are not a substitute for the full foliation-to-graph argument in M282. We use its range identification and improve the rational supremum constant by centered finite averaging in Section 6.

Volume 21 [15], `v21c01:physical-gram-definition` and `v21c01:residual-decomposition`, identifies the physical norm and the omitted component of a compressed residual. Section 7 specializes this mechanism to the rational Jacobi operator, with a proved full-space stability constant. Merely forming a positive finite Gram matrix cannot certify a continuum inverse.

The full-space contraction theorem in TR1 `stability.tex` [14], `thm:ball`, assumes a C^1 nonlinear map between stated Banach spaces and a bounded isomorphism preconditioner. Its `prop:transfer` transports a valid equation through bounded coordinate isomorphisms. These assumptions do not hold automatically for an analytic-scale composition equation with derivative loss. We do not apply that theorem to an unrestricted Wiener ball. The more limited use is its ledger of true residual, derivative defect, invariant ball and model error. Volume 21's `v21c10:seqtheorem` [16] illustrates why a complete weighted tail, rather than a zero retained residual, is essential.

Finally, TR8's `thm:coercive` [17] contains the classical Hilbert-space coercivity argument with an actual operator realization. We repeat that argument on the *rational nonresonant space*, not on the Dirichlet PDE space of that report. This yields a domain-correct extension rather than a claim that a billiard difference operator is an elliptic PDE.

4 Exact circle row preconditioning

The following calculation sharpens the generic high-row bound at the normalized circle and provides a nonasymptotic reference operator.

Theorem 4.1 (Exact multiplier and optimal inverse)**.** *For $f = 0$ in (2.5), consider the uniform q-cell stationary mesh with $b = 2\pi/q$ and fixed endpoints, $q \geq 2$. The high-row derivative of (2.6), on its locally differentiable stationary branch, is*

$$(\,\mathrm{d}C_{\pm q})_0(v) = d_q v_{\pm q}, \qquad d_q = \frac{q^3 \sin(\pi/q)}{\pi(q^2-1)} = \prod_{j=2}^{\infty} \left(1 - \frac{1}{q^2 j^2}\right). \tag{4.1}$$

In every weighted Fourier ℓ^1 space above, the diagonal operator $D_H v = (d_{|n|} v_n)_{|n|\geq N}$, $N \geq 2$, is a bounded isomorphism, and

$$0 < d_N \leq d_q < 1, \qquad \|D_H^{-1}\| = d_N^{-1}, \qquad 1 - d_q \leq \frac{\pi^2/6 - 1}{q^2}. \tag{4.2}$$

The inverse norm is optimal. At $N = 2$ it is $3\pi/8 < 33/28$.

Proof. At zero input $\phi(x) = x$ and the midpoint is the arithmetic midpoint, so $D_0(x, x+b) = 2\sin(b/2) - b$. Equal steps make the interior Euler equations vanish. Their fixed-endpoint Hessian is a nonzero constant multiple of the Dirichlet second difference: $D_{12} = \sin(b/2)/2 > 0$. It is invertible

on the $q-1$ interior nodes. Thus for each fixed q ordinary finite-dimensional implicit differentiation gives the local stationary branch for sufficiently regular small input variations. The uniform-in-q source mesh estimates are separate hypotheses; the calculation here does not establish them.

Differentiate its critical action. All interior vertex derivatives cancel by stationarity; endpoint derivatives are zero because the endpoints are fixed. Differentiating the two integrals in (2.5), their moving midpoint contributions cancel because the two integrands agree at that midpoint. The variation of $\phi(z)-\phi(X)$ is $\int_X^z v(s)\,\mathrm{d}s$. Reversing the resulting triangular integral on the left half gives

$$2(1-\cos s)v(x+s)-v(x+s)\int_s^{b/2}\sin z\,\mathrm{d}z=[2+\cos(b/2)-3\cos s]v(x+s).$$

The right half has the reflected kernel. Hence the exact first variation of one edge is (3.1). It is a translation-invariant convolution on the lifted interval, so each Fourier mode retains its frequency. Summing the q translated cells and extracting output frequency q cannot mix other input frequencies into that row.

For the q-th mode, the integral of the constant part of K_b vanishes, and symmetry, $e^{\mathrm{i}qb}=1$, gives

$$\begin{aligned}\int_0^b\cos(\min(s,b-s))e^{\mathrm{i}qs}\,\mathrm{d}s&=2\int_0^{\pi/q}\cos s\cos(qs)\,\mathrm{d}s\\&=\frac{\sin((q-1)\pi/q)}{q-1}+\frac{\sin((q+1)\pi/q)}{q+1}=\frac{2\sin(\pi/q)}{q^2-1}.\end{aligned}$$

The sum over cells contributes q, and multiplying by $-q^2/(6\pi)$ proves the first identity in (4.1). The negative row follows by conjugation. The same argument for general smooth v follows by Fourier approximation and the bounded cell integral.

Euler's sine product [10], with its $j=1$ factor removed, proves the product formula. Since $\sum_{j\ge2}(qj)^{-2}<\infty$ and each factor is strictly between zero and one, the product is positive and less than one. Each factor increases with q, giving $d_N\le d_q$. The finite inequality $1-\prod(1-a_j)\le\sum a_j$ for $0\le a_j\le1$ passes to the limit and proves the last bound in (4.2). Multiplication by a bounded diagonal sequence on weighted ℓ^1 has norm equal to its supremum modulus. A single N-th mode attains the inverse norm; a real cosine gives the same norm on the real-symmetric subspace. Finally $d_2=8/(3\pi)$ and $\pi<22/7$ gives the rational upper bound. No assertion that the source's admissible row cutoff equals two is involved. □

Corollary 4.2 (Reference-centered high-mode inverse)**.** *Suppose the source's row estimates hold along a bounded input segment from* 0 *to* f*, with enough derivatives for the chosen weight, and let*

$$E_N\ge\|A_H(f)-D_H\|_{\sigma,a},\qquad E_N<d_N.$$

Then

$$\|A_H(f)^{-1}\|_{\sigma,a}\le\frac{1}{d_N-E_N}.\tag{4.3}$$

Under (2.7) *one can take* $E_N\le C\|f\|_{r_0,\sigma}/N$*, with its source-dependent constant. For* g *in the full output space and any high-mode trial* v*,*

$$\|A_H(f)^{-1}g-v\|_{\sigma,a}\le\frac{\|g-A_H(f)v\|_{\sigma,a}}{d_N-E_N}.$$

Proof. The same column summation as (2.8), applied to the difference rows, gives the stated E_N estimate. Factor $A_H=D_H[I+D_H^{-1}(A_H-D_H)]$ and apply the Neumann series, since the

perturbation norm is at most $E_N/d_N < 1$. This is a two-sided factorization between the same complete high-mode spaces, so it proves surjectivity as well as a lower bound. Apply the inverse to the residual for the final assertion. □

The improvement is exact reference preconditioning and an explicit margin, not a claim that arbitrary normalized inputs are small. Large input bounds change C and the admissible cutoff. Nonlinear corrections still need the derivative and strip-width budget; a linear inverse bound does not suppress composition losses. Low modes $|n| < N$ are held fixed, not determined or shown to be the ellipse modes by this argument.

5 Explicit analytic-scale Newton budgets

The source's boundary-analyticity proof uses the same finite-derivative input class at every step. The following standard Newton algebra provides a sharper stopping rule once its constants have been validated. It does not introduce a new Newton method.

Theorem 5.1 (Ball closure and double-exponential tails)**.** *Let $r \geq 0$, $A, B, P > 0$, $b = 2^r$, $C = AP^r$ and $\sigma_j = (1 + 2^{-j})/(2P)$. Suppose the available Newton construction, whenever its correcting segment stays inside a fixed input class, has*

$$\varepsilon_{j+1} \leq AP^r 2^{rj} \varepsilon_j^2, \qquad \|\Delta f_j\|_{C^r(\sigma_{j+1})} \leq BP^r 2^{rj} \varepsilon_j. \tag{5.1}$$

Suppose also that the correction exists with the second bound from the current admissible state before testing the next segment. Let $m > 0$ be the available input-norm margin at the initializer. If

$$z_0 = Cb\varepsilon_0 < 1, \qquad \frac{B}{Ab} \frac{z_0}{1 - z_0} < m, \tag{5.2}$$

then every correcting segment stays in the class, and

$$\varepsilon_j \leq \frac{z_0^{2^j}}{Cb^{j+1}}, \qquad \|\Delta f_j\|_{C^r(\sigma_{j+1})} \leq \frac{B}{Ab} z_0^{2^j}. \tag{5.3}$$

There is a limit on the positive strip of width $1/(2P)$ and, after J corrections,

$$\|f_* - f_J\|_{C^r(1/(2P))} \leq \frac{B}{Ab} \frac{z_0^{2^J}}{1 - z_0^{2^J}}. \tag{5.4}$$

If the original equation is continuous in the limiting topology on its required smaller domains, and its residuals tend to zero there, the limit solves that equation.

Proof. Set $z_j = Cb^{j+1}\varepsilon_j$. On any valid step, $z_{j+1} \leq z_j^2$, hence inductively $z_j \leq z_0^{2^j}$. The correction bound in (5.1) becomes $Bz_j/(Ab)$. To avoid circular use of the recurrence, assume only the already established bound through j. The correction at that state exists by the stated construction. Its entire segment, together with the previous segments, uses at most $B(Ab)^{-1} \sum_{i=0}^{j} z_0^{2^i} \leq B(Ab)^{-1} z_0/(1 - z_0) < m$. All earlier bounds remain valid when restricted to the current smaller strip. Thus the current segment is admissible before using the quadratic residual bound. That proves the induction. For $x = z_0^{2^J} < 1$, the tail sum is $\sum_{k \geq 0} x^{2^k} \leq \sum_{l \geq 1} x^l = x/(1 - x)$. Uniform summability of the corrections and their stated derivatives on the limiting strip gives (5.4). Uniform limits on compact subsets of holomorphic functions are holomorphic. Continuity and residual convergence give the last assertion; they are included explicitly rather than inferred from a small finite list of residuals. □

For the source's $r = r_0 = 17$, sufficiently fast initial truncation decay can meet (5.2); this remains an existential statement until A, B, m, P are enclosed for the actual input. For example the abstract rational data $r = 2$, $A = B = 1$, $P = 2$, $\varepsilon_0 = 1/64$ give $z_0 = 1/4$ and total correction at most $1/12$. With input margin $m = 1/10$ the test passes. After $J = 2$, the remaining bound is $1/1020$. These are exact arithmetic fixtures, not measured constants for a noncircular table. They display what a reproducible stopping certificate would need to record. For a desired tail $\tau > 0$, it suffices to have $z_0^{2^J} \le \tau/(B/(Ab) + \tau)$.

There is a useful further elementary sharpening. For $0 \le x < 1$,

$$\sum_{j\ge 0} x^{2^j} \le x + \frac{x^2}{1-x^2} \le \frac{x}{1-x}. \tag{5.5}$$

Indeed $2^j \ge 2j$ for $j \ge 1$, so the remaining powers are bounded by $\sum_{j\ge 1} x^{2j}$. Thus one may replace the ball majorant by $B(Ab)^{-1}[z_0 + z_0^2/(1-z_0^2)]$ and the tail majorant by the same expression with z_0 replaced by $z_0^{2^J}$. The induction in the proof is unchanged. For the displayed rational fixture the total bound improves from $1/12$ to $19/240$; the tail after two corrections improves from $1/1020$ to $65791/67107840$. Both tail estimates live on the *limiting* strip, not the larger current strip.

The joint-parameter iteration in the source is different: after choosing $N_i \simeq \delta_i^{-1} \log(1/\varepsilon_i)$ it absorbs logarithms to obtain exponent $7/4$. The present squaring lemma is applied to the boundary-analyticity recurrence, not silently to that different one.

6 Rational resonance: exact and centered inverses

Fix a reduced real $t = 2\pi l/q$, $q \ge 2$. On 2π-periodic functions put $T = T_t$, $S = T_{t/2}$, and

$$\Pi = \frac{1}{q}\sum_{j=0}^{q-1} T^j, \qquad K = \ker \Pi, \qquad \partial = \frac{S - S^{-1}}{t} = \frac{S^{-1}(T-I)}{t}. \tag{6.1}$$

Here K denotes the closed subspace in whichever norm is explicitly specified. Π retains precisely Fourier indices divisible by q. In particular the excluded resonant space is infinite dimensional on the circle; excluding only the mean is insufficient. $\partial\Pi = 0$, $\Pi\partial = 0$, and S commutes with Π. The half shift need not be the identity on $\operatorname{ran}\Pi$.

Theorem 6.1 (Centered cyclic primitive)**.** *For a rotation acting on periodic L^∞ or on the Banach space of bounded periodic holomorphic functions on a fixed strip, define*

$$R_q = \frac{1}{q}\sum_{j=0}^{q-1}\left(j - \frac{q-1}{2}\right)T^j, \qquad c_q = \frac{\lfloor q^2/4 \rfloor}{q}. \tag{6.2}$$

Then $(T-I)R_q = R_q(T-I) = I - \Pi$, $\Pi R_q = 0$, and R_q is the canonical inverse of $T - I$ on K, with

$$\|R_q\|_{K\to K} \le c_q, \qquad \partial^{-1}|_K = tR_qS, \qquad \|\partial^{-1}\|_{K\to K} \le |t| c_q. \tag{6.3}$$

The $L^\infty(K)$ norm of R_q is exactly c_q. The strip estimate is an upper bound; no strip-norm extremizer assertion is made.

Proof. Telescoping the finite polynomial gives $(T-I)q^{-1}\sum jT^j = I-\Pi$. The centered term is a multiple of Π and is annihilated by $T-I$. Summing the centered weights gives zero, hence $\Pi R_q = 0$. All translations are isometries for the specified norms, and

$$\frac{1}{q}\sum_{j=0}^{q-1}\left|j-\frac{q-1}{2}\right| = \begin{cases} q/4, & q \text{ even}, \\ (q^2-1)/(4q), & q \text{ odd}. \end{cases}$$

This proves the norm bound. Algebra in (6.1), together with commutation, gives both inverse identities for tR_qS on K.

For sharpness take $g_j = \operatorname{sgn}(j-(q-1)/2)$ on a q-point orbit, with middle value zero when q is odd. Then $\sum g_j = 0$, $\|g\|_\infty = 1$, and at the selected initial point $(R_qg)_0 = q^{-1}\sum|j-(q-1)/2| = c_q$. To realize this in L^∞ on the circle, use a measurable fundamental interval for the cyclic rotation and assign the same orbit pattern over it; its finitely many translates partition the circle modulo endpoints. The equality then holds on a set of positive measure, not just at a single null-set point. For continuous functions one may instead multiply this pattern by a continuous bump supported inside that fundamental interval and extend around its disjoint translates. The same values are attained on a smaller interval. Holomorphic strip norms introduce additional constraints, so only the already proved upper bound is used there. □

The uncentered coefficient estimate $q^{-1}\sum j = (q-1)/2$ is larger for $q > 2$. Centering is legitimate because $\Pi g = 0$; minimizing the sum of absolute coefficients amounts to choosing a median of $0,\dots,q-1$. The improvement is classical finite cyclic algebra, specialized to the source's normalized difference operator. It does not remove resonant compatibility.

Corollary 6.2 (An explicit strip-supremum Jacobi bound)**.** *On a fixed strip at a fixed* real *rational parameter, suppose a is bounded and holomorphic and $\delta_a = \|a-1/4\|_\infty < 1/8$. The operator $\mathcal{L} = \partial a\partial : K \to K$ is invertible and*

$$\|\mathcal{L}^{-1}\| \le \frac{4t^2c_q^2}{1-8\delta_a}. \tag{6.4}$$

Proof. Let $M = (I-\Pi)a|_K$. Since $\|\Pi\| \le 1$, $\|M-(1/4)I\| \le 2\delta_a$ and $\|M^{-1}\| \le 4/(1-8\delta_a)$ by a Neumann series. The factorization $\mathcal{L} = \partial M\partial$ is exact because $\partial\Pi = 0$. Its inverse is $\partial^{-1}M^{-1}\partial^{-1}$, and Theorem 6.1 proves (6.4). Each factor preserves K and is a two-sided inverse there. □

This gives explicit constants for the cyclic inversion inside `joint:compatibility`. The source still needs shrinking strips for derivatives of the assigned jets, bounds on the changing coefficient, and polynomial smallness in N and the buffer. Real translations are isometries on these strips; complex parameter shifts are not. The corollary is therefore not a substitute for buffered two-variable division.

At fixed q the projection itself gives a slightly better sufficient test. Since

$$I-\Pi = (1-1/q)I - q^{-1}\sum_{j=1}^{q-1}T^j, \qquad \|I-\Pi\| \le 2(1-1/q),$$

the same proof replaces (6.4) by

$$\|\mathcal{L}^{-1}\| \le \frac{4t^2c_q^2}{1-8(1-1/q)\delta_a}, \quad 8(1-1/q)\delta_a < 1. \tag{6.5}$$

This bounds the composed inverse; it does not assert that its constant is optimal for a variable coefficient.

6.1 An explicit reciprocal-average flux inverse

The preceding near-constant test is sufficient but not necessary. The finite cyclic structure gives a more informative exact inverse formula, with no derivative loss in y at a fixed real rational parameter.

Theorem 6.3 (Orbitwise flux certificate)**.** *Work either in periodic L^∞ or bounded periodic holomorphic functions on a fixed strip. Let t be as in* (6.1) *and $R = \partial^{-1}|_K$, $\|R\| \le k_t := |t|c_q$. Suppose a is bounded, $b = 1/a$ belongs to the same bounded function algebra, and*

$$\|b\| \le B, \qquad |\Pi b| \ge \beta > 0 \tag{6.6}$$

throughout the domain (almost everywhere for L^∞). Then $\mathcal{L} = \partial a \partial : K \to K$ is a bounded isomorphism. For $f \in K$ its solution is

$$H = Rf, \quad h = -\frac{\Pi(bH)}{\Pi b}, \quad w = b(H + h), \quad v = Rw. \tag{6.7}$$

For any constant $b_0 \in \mathbb{C}$, put $\omega = \|b - b_0\|$. Then

$$\|\mathcal{L}^{-1}\| \le k_t^2 B(1 + \omega/\beta). \tag{6.8}$$

Taking $b_0 = 0$ gives $k_t^2 B(1 + B/\beta)$. For constant a, choosing $b_0 = b$ gives the sharper bound $k_t^2|b|$. On the real L^∞ domain, real $a \ge \alpha > 0$ additionally gives $\|\mathcal{L}^{-1}\| \le 2k_t^2/\alpha$.

Proof. The quotient h is invariant under T: both its numerator and denominator are in $\operatorname{ran}\Pi$. For any such invariant function, $\Pi(bh) = h\Pi b$. Hence $\Pi w = \Pi(bH) + h\Pi b = 0$. Thus Rw is defined, $\partial v = w$, and $a\partial v = H + h$. Since $\partial h = 0$ and $\partial H = f$, the formula gives $\mathcal{L}v = f$.

If $\mathcal{L}v = 0$ for $v \in K$, then $a\partial v = h$ for some $h \in \operatorname{ran}\Pi$, because the kernel of ∂ is exactly $\operatorname{ran}\Pi$. Therefore $\partial v = bh$, and applying Π yields $0 = h\Pi b$. The uniform nonvanishing assumption forces $h = 0$, then $v = 0$. This proves uniqueness and a two-sided inverse.

Since $\Pi H = 0$, the numerator of h is $\Pi((b - b_0)H)$, so $\|h\| \le \omega\|H\|/\beta$. Consequently $\|w\| \le B(1 + \omega/\beta)\|H\|$; two uses of the centered primitive give (6.8). All operations preserve bounded holomorphy on the same strip: translations are real, products remain holomorphic, and the denominator is uniformly nonzero. In the real positive case, $-h$ is the positive b-weighted mean of H along the finite orbit, so $|h| \le \|H\|$ even for complex-valued H. Together with $B \le 1/\alpha$ this gives the last bound. Here the denominator lower bound follows as well from boundedness of positive a: $\Pi(1/a) \ge 1/\|a\|_\infty$. □

This is an exact cyclic flux, or Schur-complement, calculation. Its denominator is an *orbit-invariant function*, not just the scalar global mean. Nonvanishing of a alone is insufficient. For $q = 2$, $t = \pi$, let $a(y) = e^{-\mathrm{i}y}$, so $b = e^{\mathrm{i}y}$ and $\Pi b = 0$. Then $w = b \in K$, $v = Rw \ne 0$, but $\partial a \partial v = \partial 1 = 0$. This is a bounded holomorphic counterexample on every fixed strip to omitting the reciprocal-average gate. It is not a positive real billiard coefficient or a counterexample to the source theorem.

The flux certificate weakens the near-$1/4$ condition at the fixed rational strip step, often substantially. Uniform bounds as t varies, derivatives of its reciprocal average, rational-center compatibility, and the joint complex-domain construction still require their own validation. The real L^2 theorem next supplies a different, sometimes sharper metric estimate; neither theorem subsumes all norms of the other.

Corollary 6.4 (A wider complex-strip coefficient disk)**.** *In the fixed-strip setting of Theorem 6.3, let $c > 0$ be constant and $\|a - c\| \le \delta < c$. Then*

$$\|\mathcal{L}^{-1}\| \le \frac{k_t^2 c}{(c-\delta)^2}. \tag{6.9}$$

In particular, for $c = 1/4$ the condition is $\delta < 1/4$ and the bound is $4k_t^2/(1-4\delta)^2$.

Proof. Writing $b = 1/a$ and squaring $|1 - cb| \le \delta|b|$ gives the reciprocal disk

$$\left| b - \frac{c}{c^2 - \delta^2} \right| \le \frac{\delta}{c^2 - \delta^2}.$$

Thus $B \le 1/(c-\delta)$ and $\operatorname{Re} \Pi b \ge 1/(c+\delta)$. With $b_0 = c/(c^2-\delta^2)$ and $\omega = \delta/(c^2-\delta^2)$, Theorem 6.3 gives $k_t^2(c-\delta)^{-1}(1 + \delta/(c-\delta))$, which is (6.9). All inequalities hold throughout the complex strip; they do not infer complex positivity merely from real-axis values. □

For $0 < \delta < 1/8$, $(1-4\delta)^2 > 1 - 8\delta$, so the last bound improves the unrefined Neumann estimate as well as allowing a wider coefficient disk. The finite-q bound (6.5) can be better for small perturbations; one may take the minimum of the valid bounds. The general reciprocal-average test need not require proximity to any constant.

6.2 A real coercive extension beyond near-constant coefficients

Use normalized measure $\mathrm{d}y/(2\pi)$ on $\mathbb{T}$, and the Hilbert inner product conjugate-linear in its first argument. The following operator is bounded on all of its stated $K \subset L^2$, since translations are bounded: no unmentioned Sobolev domain is needed at fixed $t \neq 0$.

Theorem 6.5 (Sharp universal rational coercivity bound)**.** *Let $a \in L^\infty(\mathbb{T}; \mathbb{C})$ with $\operatorname{Re} a \ge \alpha > 0$ almost everywhere. On $K = \ker \Pi \subset L^2(\mathbb{T})$, the operator $A = -\partial a \partial$ is a bounded isomorphism and*

$$\|A^{-1}\|_{2\to 2} \le \frac{t^2}{4\alpha \sin^2(\pi/q)}. \tag{6.10}$$

This is optimal as a bound depending only on α, t, q: equality holds for $a \equiv \alpha$. For nonconstant a, it is an upper bound, not a formula for the exact norm.

Proof. $\partial^* = -\partial$. Its Fourier multipliers are $2\mathrm{i}\sin(nt/2)/t$. For indices not divisible by q, reducedness of l/q permutes the nonzero residues modulo q, so

$$\|\partial w\|_2 \ge d_* \|w\|_2, \qquad d_* = 2\sin(\pi/q)/|t|, \quad w \in K. \tag{6.11}$$

Parseval proves this on the full closed subspace; a least-residue Fourier mode attains equality. Since $\Pi\partial = 0$, A maps K into itself. Integration by the bounded adjoint identity gives

$$\operatorname{Re}\langle w, Aw\rangle = \int \operatorname{Re} a \, |\partial w|^2 \frac{\mathrm{d}y}{2\pi} \ge \alpha d_*^2 \|w\|_2^2.$$

Consequently $\|Aw\|_2 \ge \alpha d_*^2 \|w\|_2$, giving injectivity and closed range. If $v \in K$ is orthogonal to the range, then $\langle v, Au\rangle = 0$ for all $u \in K$. Choosing $u = v$ and taking the real part forces $v = 0$. The range is therefore dense and closed, hence all of K. The inverse bound follows. For constant $a = \alpha$ the Fourier eigenvalues are $\alpha|d_n(t)|^2$, and a least-residue mode attains the stated norm. □

This is the classical coercive-form argument underlying Lax–Milgram, in the correct rational difference space. It replaces near-constantness by accretivity for this *real-parameter L^2 step*. For example $a(y) = 1 + \frac{1}{2}\cos y$ has $\alpha = 1/2$ although $\|a - 1/4\|_\infty = 5/4$, far outside Corollary 6.2. Neither real measurability nor positivity provides holomorphic-strip regularity, estimates for a_t, or simultaneous complex-domain control. The source's full joint analytic theorem cannot be weakened to these real hypotheses by this corollary alone. The Fourier proof also shows why retaining the whole resonant projection is necessary: a nonzero q-th harmonic lies in the kernel of ∂, regardless of a.

6.3 Why first jets carry additional information

At a simple zero t_0 of a scalar divisor d, the identity $d(t)u(t) = F(t)$ with holomorphic u requires $F(t_0) = 0$, but it then sets $u(t_0) = F'(t_0)/d'(t_0)$. Thus two forcing functions with the same zero value can produce arbitrarily different primitive values. For example $F(t) = Md(t)$ has zero resonant value for every M, while the primitive equals M. This elementary fact is the precise reason the first compatibility alone does not close the source's second division. It is not evidence of a defect in M282: that manuscript explicitly constructs and uses the first jets. The improved inverses above sharpen that construction's constants; they do not replace its jet argument or imply that independently assigned jets glue to an exact family without the subsequent analytic iteration.

7 Physical Gram and complete rational residuals

Let A and K be as in Theorem 6.5, and write $\gamma = \alpha d_*^2$. For a finite injective synthesis $J : \mathbb{C}^m \to K$, define

$$G = J^*J > 0, \qquad P_J = JG^{-1}J^*, \qquad H = J^*AJ. \tag{7.1}$$

The synthesis functions may be nonorthogonal Ram atoms; membership in K must be checked or enforced, for example by applying $I - \Pi$ and then removing dependencies. Gram positivity only on that finite space does not prove the range condition by itself.

Proposition 7.1 (Coercive metric and full-residual certificate)**.** *One has* $\mathrm{Re}(c^*Hc) \geq \gamma c^*Gc$ *for every coefficient vector. For* $f \in K$, $u = A^{-1}f$, $u_c = Jc$, *and* $r = f - AJc$, *put* $b = J^*r$. *Then*

$$\|r\|_2^2 = b^*G^{-1}b + \|(I - P_J)r\|_2^2, \qquad \|u - u_c\|_2 \leq \gamma^{-1}\sqrt{b^*G^{-1}b + \|(I - P_J)r\|_2^2}. \tag{7.2}$$

If a is real, $H = H^*$; in any event its Hermitian part satisfies $(H + H^*)/2 \succeq \gamma G$.

Proof. Insert Jc in the full-space coercivity inequality. Since P_J is the orthogonal projection onto $\operatorname{ran} J$, its residual component is $JG^{-1}b$, with squared norm $b^*G^{-1}b$. Pythagoras gives the identity and Theorem 6.5 gives the error bound. For real a, the bounded operator $-\partial a\partial$ is self-adjoint, so its finite compression is Hermitian. The accretive case follows by taking real parts of quadratic forms. □

For a Fourier synthesis with nonresonant indices n_j, $G = I$ and

$$H_{jk} = \overline{d_{n_j}(t)}\, a_{n_j - n_k}\, d_{n_k}(t). \tag{7.3}$$

This is a useful direct assembly identity. It retains complex conjugation and the exact normalized measure. If a and f are trigonometric polynomials, the full residual for finite Jc is another finite

trigonometric polynomial, so every omitted coefficient can be computed. For an analytic coefficient with a verified weighted tail, the same identity separates retained products and the full convolution tail. Without such information the last term in (7.2) is unknown, not zero.

If only an approximate Gram matrix is available, bounds must be relative: $a_0 G \preceq \widehat{G} \preceq b_0 G$, $a_0 > 0$, give norm conversion. Near a nearly dependent synthesis, a small absolute entry error is not enough. This is exactly the scope of Volume 21's physical-metric propositions, now supplied with a concrete M282 operator and a proved inverse. A finite residual of zero can still conceal components generated by variable-coefficient multiplication outside the chosen Fourier bank. Nor may the L^2 norm in (7.2) be substituted for the holomorphic supremum norm used in the joint Newton proof without a separately quantified regularity estimate.

8 Quantitative physical collars

The source's final support-envelope step admits effective margins once its analytic input is known. The following theorem is geometric. Its caustic conclusion is deliberately an additional implication.

Theorem 8.1 (A quantitative support-collar certificate)**.** *Let $g(\theta, \lambda)$ be real, 2π-periodic in θ, and C^2 on a neighborhood of $\mathbb{T} \times [0, L]$. Suppose throughout that cylinder*

$$g \geq h_* > 0, \qquad 0 < \rho_* \leq \rho := g + g_{\theta\theta} \leq \rho^*, \qquad 0 < c_* \leq -g_\lambda \leq c^*. \tag{8.1}$$

Let $K_\lambda = \bigcap_\theta \{z : z \cdot n_\theta \leq g(\theta, \lambda)\}$ and $\Gamma(\theta, \lambda) = g n_\theta + g_\theta e_\theta$. Then $\Gamma(\cdot, \lambda)$ is the simple regular strictly convex boundary of K_λ, with curvature $1/\rho$; the bodies are strictly nested. The map on $\mathbb{T} \times [0, L)$ is a homeomorphism onto $K_0 \setminus K_L$. Moreover:

$$\begin{aligned} K_\mu + c_*(\mu - \lambda)\overline{B} &\subset K_\lambda && (0 \leq \lambda < \mu \leq L), \\ \det(\Gamma_\theta, \Gamma_\lambda) &= -\rho g_\lambda \geq \rho_* c_*, \\ |\lambda(z) - \lambda(w)| &\leq |z - w|/c_*, \\ 2\pi\rho_* c_*(\mu - \lambda) \leq |K_\lambda| - |K_\mu| &\leq 2\pi\rho^* c^*(\mu - \lambda). \end{aligned} \tag{8.2}$$

Here $\lambda(z)$ is the collar coordinate and $\overline{B}$ the closed unit disk. If g is jointly real analytic near the closed cylinder, so is Γ, and its local inverse is analytic there. If, additionally, the oriented line graph $p = g(\theta, \lambda)$ is mapped onto itself by the actual billiard of K_0 for each positive λ, then the positive curves are caustics at both reflection endpoints and in both orientations.

Proof. The positive lower support bound puts a disk inside every K_λ; the finite upper bound makes it compact. Differentiating gives $\Gamma_\theta = \rho e_\theta$. For fixed α,

$$\frac{\mathrm{d}}{\mathrm{d}\theta}(\Gamma(\theta, \lambda) \cdot n_\alpha) = \rho(\theta, \lambda) \sin(\alpha - \theta).$$

Over one circle this has its unique maximum at $\theta = \alpha$, of value $g(\alpha, \lambda)$. Thus all Γ points satisfy all support inequalities, with a unique contact for each normal. Conversely a boundary point z must have a zero of the nonnegative support slack $g - z \cdot n$; otherwise its strictly positive minimum would give an interior disk around z. At that zero the angular derivative also vanishes, yielding $z \cdot n = g$, $z \cdot e = g_\theta$, hence $z = \Gamma$. The unique-maximum property proves injectivity on each circle, and $\rho > 0$ proves regularity and the curvature formula.

Integration of $g_\lambda \leq -c_*$ gives $g(\theta, \mu) + c_*(\mu - \lambda) \leq g(\theta, \lambda)$. For $x \in K_\mu$ and $|v| \leq c_*(\mu - \lambda)$, every support inequality holds for $x + v$, proving the first inclusion. Thus distinct boundaries are disjoint.

The continuous map on the compact closed cylinder is injective and therefore a homeomorphism onto its image.

For $z \in K_0 \setminus K_L$, define $m_z(\lambda) = \min_\theta(g(\theta,\lambda) - z\cdot n_\theta)$. It is continuous, nonnegative at zero, negative at L, and $m_z(\mu) \le m_z(\lambda) - c_*(\mu-\lambda)$. It has exactly one zero. At that zero z is on exactly one support curve, proving the claimed image of the half-open cylinder. Also $|m_z(\lambda) - m_w(\lambda)| \le |z-w|$. Comparing the two zero parameters with strict decrease gives the global Lipschitz bound for $\lambda(z)$ in (8.2), even though the collar itself need not be convex.

Now $\Gamma_\lambda = g_\lambda n + g_{\theta\lambda} e$ and $\det(e,n) = -1$, giving the determinant formula. Its nonvanishing gives the asserted local inverse, including extension across parameter endpoints under the analytic hypothesis. Its gradient satisfies $\nabla\lambda = n/g_\lambda$, consistent with the previous norm bound. The area formula and periodic integration by parts give

$$|K_\lambda| = \frac{1}{2}\int_0^{2\pi} (g^2 - g_\theta^2)\,\mathrm{d}\theta, \quad \frac{\mathrm{d}}{\mathrm{d}\lambda}|K_\lambda| = \int_0^{2\pi} g_\lambda \rho\,\mathrm{d}\theta. \tag{8.3}$$

Integration of the upper and lower bounds for $-g_\lambda\rho$ proves the area bounds. Equivalently, they follow from the nonvanishing collar Jacobian. The perimeter is $\int \rho = \int g$, so its loss between two parameters lies between $2\pi c_*(\mu-\lambda)$ and $2\pi c^*(\mu-\lambda)$.

For the last assertion the tangent at $\Gamma(\theta,\lambda)$ is exactly the oriented line state $(\theta, g(\theta,\lambda))$. Its contact point lies in $\operatorname{int} K_0$ for $\lambda > 0$, so the line meets the table. Onto invariance supplies both the successor and predecessor tangent segments. Reflection is an involution and commutes with reversal of the ordered incoming/outgoing pair; therefore the opposite orientation also preserves tangency. This proves precisely the physical caustic property, not only local tangency of a formal envelope. □

8.1 An explicit admissible width from the source's expansion

Suppose $g = h - c\lambda + e$, $e(\theta,0) = 0$, with verified bounds

$$\|e\|_\infty \le M_0\lambda^2, \quad \|e_{\theta\theta}\|_\infty \le M_2\lambda^2, \quad \|e_\lambda\|_\infty \le M_\lambda\lambda \quad (0 \le \lambda \le L). \tag{8.4}$$

Let $h_0 = \min h$, $r_0 = \min(h + h'')$, $c_0 = \min c > 0$, $C_0 = \|c\|_\infty$ and $C_2 = \|c + c''\|_\infty$. The three explicit strict inequalities

$$h_0 - C_0 L - M_0 L^2 > 0, \quad r_0 - C_2 L - (M_0 + M_2)L^2 > 0, \quad c_0 - M_\lambda L > 0 \tag{8.5}$$

supply h_*, ρ_*, c_* in Theorem 8.1. Upper constants come from the corresponding plus signs. This follows simply by bounding the three displayed quantities; it is an all-cylinder estimate, not a grid test. Analyticity alone gives qualitative finite bounds after shrinking the domain, but does not supply their numerical enclosures. If the combined remainder is directly bounded by $\|e + e_{\theta\theta}\|_\infty \le M_\rho\lambda^2$, then M_ρ can replace $M_0 + M_2$ in the curvature test, retaining any cancellation certified in that combined quantity.

For the actual source, (2.12) identifies the positive leading coefficient in terms of the original curvature radius. In particular the leading area and perimeter losses are

$$\begin{aligned} |K_0| - |K_\lambda| &= \frac{\lambda}{8k^2}\int_0^{2\pi} r^{4/3}\,\mathrm{d}\theta + O(\lambda^2), \\ \operatorname{Per}(K_0) - \operatorname{Per}(K_\lambda) &= \frac{\lambda}{8k^2}\int_0^{2\pi} r^{1/3}\,\mathrm{d}\theta + O(\lambda^2) = \frac{\pi\lambda}{4k^3} + O(\lambda^2). \end{aligned} \tag{8.6}$$

These formulas use the source's normalized translation parameter; they are not invariant under an arbitrary rescaling of that parameter. They follow from (8.3) and $g_\lambda(\theta,0) = -c(\theta)$. They quantify a physical output once the exact analytic input is known, without invoking the final ellipse theorem.

8.2 Representation errors and full coefficient tails

If $\widehat{g}$ is an explicitly represented candidate and the true g satisfies uniform error bounds $\|g-\widehat{g}\|_\infty \le \eta_0$, $\|g_{\theta\theta}-\widehat{g}_{\theta\theta}\|_\infty \le \eta_2$, and $\|g_\lambda-\widehat{g}_\lambda\|_\infty \le \eta_\lambda$, then valid margins are obtained by subtracting respectively η_0, $\eta_0+\eta_2$, and η_λ from candidate margins. This includes rounding, model error, and *all* omitted coefficients.

For example, an absolutely convergent Fourier–power representation $g=\sum_{n\in\mathbb{Z},k\ge0} a_{nk}e^{in\theta}\lambda^k$ with verified differentiated absolute sums gives the tail bounds

$$\eta_{i,j} \le \sum_{(n,k)\notin\mathcal{I},\ k\ge j} |n|^i (k)_j |a_{nk}| L^{k-j}, \qquad (i,j)=(0,0),(2,0),(0,1), \tag{8.7}$$

where $(k)_j = k!/(k-j)!$ and $|n|^0=1$. The finite retained index set $\mathcal{I}$ is not itself a tail certificate. These sums are sufficient to justify differentiated convergence on the real cylinder; analytic extension to a larger complex cylinder is a separate weighted-sum condition. This is the same full-model principle as the local Ram remainder and sequence-space sources, specialized to physically measurable curvature and collar width.

8.3 Exact positive and adverse controls

Example 8.2 (The disk realizes all geometric and dynamic conditions)**.** For a disk of radius R, $r=R$, $k=R^{-1/3}$, $u=0$ and $V(y,t)=y$. The true line graph and support envelope are

$$g(\theta,\lambda)=R\cos(\sqrt{\lambda}/2), \qquad \Gamma=gn_\theta, \qquad \lambda=t^2.$$

The cosine power series is an entire function of λ. The disk line map preserves p and advances θ by $2\arccos(p/R)$, so each graph is exactly invariant with translation t. For $0\le\lambda\le1$,

$$g=\rho\ge 7R/8, \qquad -g_\lambda=\frac{R}{8}\operatorname{sinc}(\sqrt{\lambda}/2)\ge 23R/192.$$

The inequalities follow from $\cos x\ge 1-x^2/2$ and $\sin x\ge x-x^3/6$ on $0\le x\le 1/2$, with removable values at zero. The exact area loss is $\pi R^2\sin^2(\sqrt{\lambda}/2)$, whose leading coefficient $\pi R^2/4$ agrees with (8.6). This is a fully realized test case, not just a formal initializer.

Example 8.3 (Analytic nested geometry is not billiard invariance)**.** Inside the unit disk, take $0\le\lambda\le 1/4$ and

$$g(\theta,\lambda)=1-\lambda/8+(\lambda/100)\cos 3\theta. \tag{8.8}$$

For the entire cylinder, exact bounds give

$$g\ge 773/800, \quad g+g''\ge 759/800, \quad 23/200\le -g_\lambda\le 27/200.$$

Thus Theorem 8.1 supplies a full analytic nested support collar. Its exact area is $\pi[(1-\lambda/8)^2-4(\lambda/100)^2]$. For positive λ this line graph is not invariant under the unit disk billiard.

Indeed an invariant continuous line graph $p=g(\theta)$ in $-1<p<1$ for that billiard would satisfy constancy of g along every forward orbit with angular advance $2\arccos g(\theta)$. If one advance divided by 2π is irrational, the orbit is dense and continuity makes g constant. Otherwise the continuous function $\arccos(g(\theta))/\pi$ takes values only in the rationals; its connected image is a singleton, again making g constant. Equation (8.8) is nonconstant for $\lambda>0$ and lies strictly below the unit support, so it cannot be such an invariant graph. The example refutes only "analytic nested support curves imply caustics." It is not a counterexample to the source's fully invariant phase-foliation hypothesis.

9 Primary literature and exact comparison

The public-source refresh was performed on October 7, 2026 using public titles and identifiers, never confidential corpus text. These are statement and selected-proof checks, not full independent audits of the literature. The distinction between publication date and the present preprint claim is maintained.

Graph and periodic-curve regularity. Arnaud's 2009 paper [3] concerns invariant curves of exact symplectic twist maps. Its primary article and introductory formulation of Birkhoff graph regularity were consulted, not every regularity proof. Fierobe–Sorrentino [4], published in 2026, defines a periodic graph by all-point periodicity and proves regularity/invariance results in Proposition 3.1 and Appendix Proposition A.1. The latter starts with a periodic Lipschitz billiard graph; it does not assert that a rational rotation number alone gives all-point periodicity. Its analytic-family theorem concerns a fixed rational type across an already analytic family, not the smooth-to-analytic upgrade of M282.

Restricted closed-billiard rigidity. Bialy–Mironov [5], Theorem 1.1, assumes central symmetry and a foliated annulus reaching the four-periodic invariant curve. Neither condition is provided by an arbitrarily thin nonsymmetric grazing band. Kaloshin–Sorrentino [6] establishes local rigidity near ellipses; the primary journal abstract was checked. Koval's published Theorem 1 [7] treats q_0-rationally integrable domains close to an ellipse outside a locally finite exceptional eccentricity set, with $k \geq 39$, a prescribed C^k bound and sufficiently small C^1 distance. Definitions 1.1–1.3, Theorem 1, and the proof outline were read. Its dependence on the reference ellipse and on q_0 is not a uniform theorem for arbitrary positively curved tables. No such restrictions are silently added to or removed from M282's target.

Cantor and open-curve foliations. Lazutkin's original publication record and abstract [8] were checked, not its complete proof. The near-boundary invariant-curve conclusion does not supply a full continuous open-band foliation. Glutsyuk's 2024 primary article [9] treats strictly convex open curves and immersed foliations; its scope and the phase-to-caustic discussion were consulted. Open or immersed leaves must not be substituted for the simple closed global collar in M282. Flat differences of local smooth foliations also explain why equality of jets alone is not global analytic uniqueness.

Status of the new source claim. M282 and its companion [1, 2] are the public preprints being assessed, not confidential private reports. Their claims go beyond the restricted published statements just listed. The bounded comparison does not independently establish acceptance of their global conclusions, nor does the older literature refute them. We have not located or relied on a separate published proof of the unrestricted target. Euler's sine product used in our local multiplier calculation is the classical identity documented in DLMF 4.22.1 [10], not a new Ram identity.

10 Advantages, limitations, and research obligations

10.1 What is genuinely improved

The exact circle diagonal replaces a coarse near-identity estimate by the optimal inverse constant for that reference operator, retains the full high-frequency space, and makes the perturbation margin $d_N - E_N$ explicit. The centered cyclic coefficients improve a concrete strip-compatible primitive bound without changing any resonance compatibility hypothesis. The accretive L^2 inverse removes near-constantness at a specifically declared real-parameter step and supplies an exact universal constant. The orbitwise flux formula also removes near-constantness on a fixed holomorphic strip when the reciprocal and its orbit average are uniformly nonzero; its reciprocal disk corollary doubles the elementary near-1/4 admissible radius. The physical Gram specialization connects that inverse to a computable full-residual error estimate.

The analytic-scale squaring argument exposes the actual ball margin and double-exponential tail, giving an explicit stopping rule after constants are enclosed. The support-collar certificate converts an exact analytic solution and complete remainder bounds into a width, curvature, inverse coordinate and area guarantee. These gains are useful even if the global rigidity theorem is left outside the independently verified scope. No historical priority or exclusive advantage over conventional Fourier, coercive-form, convex-geometry or validated-numerics methods is claimed.

10.2 Costs and failure modes

The Fourier row bounds still depend on high finite derivatives. Larger cutoffs and narrower strips can improve a high-mode inverse while making derivative and truncation budgets more demanding. A different coordinate system cannot eliminate a true small divisor or an omitted range constraint. The circle preconditioner says nothing about the remaining low modes or a global basin of normalized inputs.

At rational parameters a full projection, not only a mean constraint, is required. The L^2 coercive extension cannot control complex parameter jets without additional analytic hypotheses. A strip inverse at isolated real rationals does not itself glue an analytic parameter family. The source's value-and-jet construction and buffered sinc divisions address exactly those extra obligations; their complexity is not removed by improving a constant in a cyclic inverse.

A small residual for a represented map must include the error between that map and the true billiard. In TR-item-336's notation, $F(u) = T_2(s, u(s)) - u(T_1(s, u(s)))$ has map uncertainty at most $\varepsilon_2 + \|u'\|_\infty \varepsilon_1$. This budget is distinct from coefficient truncation and from invariant graph existence. Likewise Example 8.3 shows that even an exact geometric collar need not be dynamically invariant. An inconclusive margin, such as $E_N \geq d_N$, is failure of a sufficient certificate, not evidence that an inverse or collar does not exist.

10.3 Concrete next investigations within M282

First, enclose the source's arbitrary-input mesh and row constants on a specified analytic input class; compute both reference-centered preconditioning and the original near-identity margin. This would turn an existential cutoff into an actual reproducible certificate for that class, without pretending to settle all tables at once.

Second, combine the centered strip inverse with the rational Dirichlet flux estimates and all fixed

Cauchy losses to obtain a completely numerical version of the joint Newton exponent and constant. Every resonant center, first jet, interpolation remainder and omitted mode must enter that bound. Real coercivity is a promising alternative only if an accompanying strip/parameter regularity theorem is proved.

Third, propagate the analytic correction tail into the support-function errors of (8.7), then use (8.5) to obtain certified physical outputs. This needs quantitative control of the angular inverse and of the composition $\phi(V)$; a small coefficient error before this composition is not automatically a small curvature error afterwards.

Finally, any independent verification of the headline ellipse claim must audit the companion's much larger global rigidity chain, beyond its three analytic inputs and final statement read here. That work is not replaced by these local improvements, by a successful circle test, or by the absence of a detected error in a bounded source read.

11 Provenance, reading depth, and verification

11.1 Public pinned mathematical sources

The catalogue row for M282 was read from `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/00-Applicability-Catalogue/applicability.json`. Its source pin is `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The bibliographic links [1, 2] use that commit, not a moving branch. The local files were read as supplied. The byte hashes recorded with the local checks identify those files; they are not a new authentication against remote Git objects.

The exact local M282 build directory is

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/Continuous-Phase-Foliations-Create-Analytic-Caustic-Collars-September-24-2026/build
```

All paths in the following table append directly to that directory. The listed files were read in their entirety, not merely searched for keywords. The section labels identify source statements and are not labels of this report.

File; lines	Main source locators and reading scope
`main.tex`; 63	Complete input list, source author/date, abstract, and bibliography setup.
`sections/introduction.tex`; 205	`thm:main`, `thm:collar-regularity`, `thm:nb-rigidity`; exact target, literature restrictions, and external final implication.
`sections/geometry.tex`; 429	`geo:support`, `geo:twists`, `geo:generating`, `geo:uniform-collar`, `geo:translation-bound`, `geo:barrier`, `prop:graph-collar`, `lem:stationary-closure`; statements and complete written proofs read.
`sections/analytic.tex`; 395	`ana:normalization`, `eq:normalized-action`, `prop:mesh-input`, `prop:row-input`, `ana:wiener`, `ana:initial-error`, `ana:glancing-interface`; arbitrary inputs, full columns, shrinking strips, and real uniqueness read.

File; lines	Main source locators and reading scope
sections/joint.tex; 730	thm:joint, joint:factorization, joint:compatibility, joint:division; all finite rational, first-jet, buffered-division, initialization and convergence arguments read.
sections/caustics.tex; 262	Complete support-envelope proof, evenness and squared parameter, actual onto invariance, positivity, nesting and relative collar coverage.
references.bib; 184	Complete bibliography read; selected current primary items refreshed as described in Section 9.
figures/phase-to-caustic.tex; 53	Complete diagram source read; not used as proof of the depicted implications.

The exact local companion build directory is

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/Rigidity-of-Smooth-Billiards-with-a-Continuous-Caustic-Collar-September-24-2026/build
```

Only the following targeted passages were read directly: sections/01-introduction.tex, lines 1–50, including Definition 1.1 and Theorem 1.2; sections/02-analytic.tex, lines 248–642, containing the stationary-mesh lemma ana:mesh-lemma (Lemma 2.3) and prop:high-frequency-estimate (Proposition 2.4), including the variation kernels and row formula; and sections/03-joint.tex, lines 436–543, containing joint:fourier-division (Lemma 3.5) and its proof. These are substantial dependency reads, not a reading of the entire companion. In particular, its later global continuation, shape, critical-curve, model and sign-test sections were not independently audited here.

11.2 Exact private sources and actual use

The private sources are local research materials, not public references uploaded or searched online. The paths below specify precisely which version was used; similar files in other directories are not implicitly included. Search maps were navigation aids only.

TR-item-336 [11]. C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-336-astra.tex. The complete mathematical text was read (1,045 lines in the inspected file). Applicable locators are eq:support, eq:curvaturebudget, thm:exclusion, eq:DF, eq:quadratic, eq:mapbudget, eq:Pq, eq:rationalinverse, eq:fulltail, thm:action, eq:actiontail, thm:uniform, thm:block, and thm:newton. Its explicit range, tail and two-sided inverse conditions are retained. Its fixed-polygon Taylor theorem is not silently applied to a moving stationary orbit, and its conditional Banach theorem is not a proof of the source's analytic-scale iteration.

Kernel coordinates and local moments [12, 13]. C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr117/TR-rev-1-old-tr117.tex, lines 238–352: eq:RT, eq:RLT, eq:IRLT, and the global/Ram equivalence discussion. C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume1-rev2/chapters/ch03-local-moments-rev2.tex, lines

1–185: thm:ram-local, source/output-centered conventions and the integrated Taylor remainder. Included subfiles were not automatically counted as read. The usable conclusion here is the finite identity (3.2) with its absolute remainder, not convergence of a smooth infinite derivative series.

Nonlinear certificates and coercive realization [14, 17]. C:/Users/mural/OneDrive/Documents/RamNonlin-IE-PDE-astra/TR1-lin-nonlin/stability.tex, lines 81–240: full derivative, thm:ball, eq:radii, prop:transfer, and the full-residual, coordinate and continuation qualifications. C:/Users/mural/OneDrive/Documents/RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex, lines 81–120: elliptic setting, thm:coercive, its proof and operator-domain statement. The original PDE domain is not identified with the rational difference space; Theorem 6.5 proves the latter's realization separately.

Physical metrics and complete tails [15, 16]. C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex, entire fragment: v21c01:physical-gram-definition, v21c01:residual-decomposition, v21c01:residual-identity, v21c01:gram-equivalence, and the finite-measurement restrictions. C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch10d-sequence.tex, entire fragment: v21c10:seqspace, v21c10:algebra, v21c10:seqtheorem, v21c10:hiddenresidual, and v21c10:residualsplit. Its model sequence equation is not asserted to be the billiard equation. The applicable mechanisms are the physical Gram norm, orthogonal residual split, and mandatory full tail.

Recursive local searches covered the four named foundations/books/IE–PDE roots and the application roots res-rt/res-math, res-rt/TechReps/FinalTechReps/TRs, RamMathUnsolved, RamPatsCore50-TRs, and RamScoutPapTRpat. The search distinguished billiard operators from unrelated optical or wave caustics. Search matches, duplicates and snapshots were not counted as theorem reads; only the actual applicable passages above ground the report. This was not an all-proof audit of the corpus.

11.3 Local computations and independent finite replay

The reproducible checker and its saved results are

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M282_report_local.py

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M282_report_local.json.

It uses exact rational arithmetic. Circle fixtures enclose π by Machin's identity and alternating arctangent series, then enclose sine by outward rational polynomial intervals. There is no binary floating-point quadrature in this checker. The finite families are:

Family	Cases
Circle interval fixtures, $2 \le q \le 128$	127
Centered half-shift inverses, reduced l/q, $2 \le q \le 30$	277
Positive variable-coefficient flux inverses	831
Nonorthogonal Gram and full-residual fixtures	90
Nonzero coefficient with zero reciprocal average	1
Analytic-scale exact recurrence steps	525
Sharper lacunary tail fixtures	16
Finite-q projection improvements	87
Complex reciprocal-disk fixtures	150
Rational support-margin fixtures	1,681
Total	3,785

The half-shift tests use $2q$ nodes, with T shifting by $2l$ and S by l. The projection therefore has two orbit classes and is *not* replaced by a single scalar mean. Exact gauges and both inverse identities are tested. The Gram tests retain the omitted residual component. The support fixtures are samples of inequalities whose all-cylinder proofs are in the text, not a claim that a grid proves positivity. The JSON also records hashes and line counts for the eight public M282 files and the checker hash.

A separate bounded proof audit was read in full from `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/M282-parent-derivations-audit.txt`. It independently checks the circle normalization, centered primitive, real coercivity, Newton indexing, and reciprocal-average flux inverse; its isometry, norm, strip and projection qualifications are incorporated above. Its recorded finite replay includes 5,049 cyclic identities, 114 positive flux fixtures, 432 Newton steps and 85 floating diagnostics, plus 376 separate rational flux fixtures with positive and signed nonzero reciprocals and nonconstant resonant flux functions. The floating diagnostics are not rigorous transcendental enclosures. Those supplied independent runs are distinguished from the separate checker executed for this report. The corresponding derivation and shared-check records are `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/batch4-parent-review.txt` and `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_batch4_parent.py` with its same-stem JSON. Their unrelated manuscript calculations are not evidence for M282.

No actionable contradiction was identified in this bounded reading of M282 and the selected imported proofs. This is a reading outcome, not a certificate of all analytic constants, all parameter domains, or the headline ellipse theorem. The independently proved conclusions of this report are precisely its displayed local theorems and conditional implications with their stated hypotheses. The remaining analytic and global obligations are those enumerated in Sections 2 and 10.

References

[1] OpenAI. *Continuous Phase Foliations Create Analytic Caustic Collars*. Public preprint, September 24, 2026. Commit-pinned source: `https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Continuous-Phase-Foliations-Create-Analytic-Caustic-Collars-September-24-2026/paper.pdf`. Local reading provenance: Section 11.

[2] OpenAI. *Rigidity of Smooth Billiards with a Continuous Caustic Collar*. Public companion preprint, September 24, 2026. Commit-pinned source: `https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Rigidity-of-Smooth-Billiards-with-a-`

Continuous-Caustic-Collar-September-24-2026/paper.pdf. Only the specific imported passages listed in Section 11 were directly read.

[3] M.-C. Arnaud. Three results on the regularity of the curves that are invariant by an exact symplectic twist map. *Publications mathématiques de l'IHÉS* **109** (2009), 1–17. https://doi.org/10.1007/s10240-009-0017-8. Primary text: https://numdam.org/item/10.1007/s10240-009-0017-8.pdf.

[4] C. Fierobe and A. Sorrentino. On the existence of periodic invariant curves for analytic families of twist maps and billiards. *Revista Matemática Iberoamericana* **42** (2026), 299–322. https://doi.org/10.4171/RMI/1595. Primary text: https://ems.press/content/serial-article-files/52360. Definition 2.6, Proposition 3.1, selected family argument, and Appendix Proposition A.1 consulted.

[5] M. Bialy and A. E. Mironov. The Birkhoff–Poritsky conjecture for centrally-symmetric billiard tables. *Annals of Mathematics* **196** (2022), 389–413. https://annals.math.princeton.edu/2022/196-1/p02. Theorem 1.1 in the author preprint: https://arxiv.org/abs/2008.03566.

[6] V. Kaloshin and A. Sorrentino. On the local Birkhoff conjecture for convex billiards. *Annals of Mathematics* **188** (2018), 315–380. https://doi.org/10.4007/annals.2018.188.1.6. Primary journal record and abstract: https://annals.math.princeton.edu/2018/188-1/p06.

[7] I. Koval. Local strong Birkhoff conjecture and local spectral uniqueness of almost every ellipse. *Inventiones mathematicae* **244** (2026), 221–298; published online December 11, 2025. https://doi.org/10.1007/s00222-025-01397-y. Definitions 1.1–1.3, Theorem 1 and introductory proof outline consulted.

[8] V. F. Lazutkin. The existence of caustics for a billiard problem in a convex domain. *Mathematics of the USSR-Izvestiya* **7** (1973), 185–214. https://doi.org/10.1070/IM1973v007n01ABEH001932. Primary publication record and abstract: https://www.mathnet.ru/eng/im2221.

[9] A. Glutsyuk. On infinitely many foliations by caustics in strictly convex open billiards. *Ergodic Theory and Dynamical Systems* **44** (2024), 1418–1467; published online July 11, 2023. https://doi.org/10.1017/etds.2023.42. Introduction, open/immersed scope, and selected Section 2.7 phase-to-envelope discussion consulted, not the entire proof.

[10] NIST Digital Library of Mathematical Functions. Section 4.22, equation 4.22.1, Euler's sine product. https://dlmf.nist.gov/4.22.E1. Consulted October 7, 2026.

[11] Local Ram corpus. TR-item-336-astra.tex. Confidential local research report on the billiard target, residuals, rational cohomology, action and inverse certificates. Exact path, labels and complete reading scope in Section 11.

[12] Local Ram corpus. Revised TR117, TR-rev-1-old-tr117.tex. Ram transforms, localization and inverse kernel coordinates. Confidential local source; exact passage and path in Section 11.

[13] Local Ram corpus. *Ram Calculus*, Volume 1, ch03-local-moments-rev2.tex, thm:ram-local. Confidential local chapter fragment; exact path and reading limits in Section 11.

[14] Local Ram corpus. TR1 linear/nonlinear equations, stability.tex, thm:ball and prop:transfer. Confidential local source; actual Banach-space assumptions and exact reading passage in Section 11.

[15] Local Ram corpus. *Ram Calculus*, Volume 21, ch01d-physical-metrics.tex. Physical Gram metrics and complete residual decomposition. Confidential local fragment; path and labels in Section 11.

[16] Local Ram corpus. *Ram Calculus*, Volume 21, ch10d-sequence.tex. Weighted sequence model, full residual and omitted tails. Confidential local fragment; path and labels in Section 11.

[17] Local Ram corpus. TR8 linear/nonlinear equations, `TR8-lin-non-lin.tex`, `thm:coercive`. Classical coercive operator realization in the stated PDE setting. Confidential local source; exact passage and path in Section 11.